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MATHEMATICS **9709/52**

Paper 5 Probability and Statistics 1 **February/March 2024**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **17** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate’s intent clear.

Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M	Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
A	Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
B	Mark for a correct result or statement independent of method marks.
DM or **DB**	When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
FT	Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

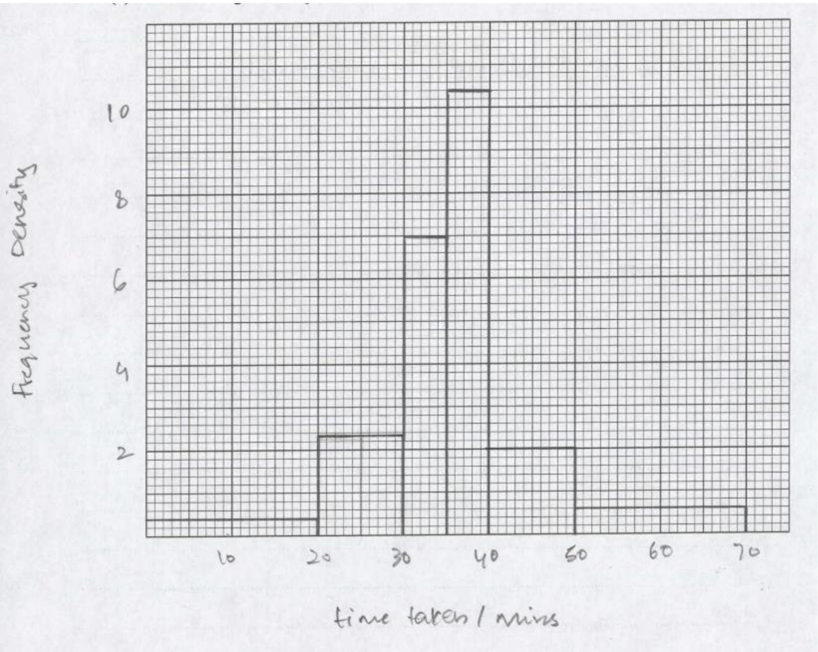
Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

Question	Answer	Marks	Guidance
1(a)	$\left[\frac{3}{12}\times\frac{2}{11}=\right]\ \frac{1}{22}$	**B1**	Accept $\frac{6}{132}$, $0.045\dot{4}\dot{5}...$ to at least three significant figures.
		1	
1(b)	$\text{P(B1}\mid\text{R2)} = \dfrac{\frac{9}{12}\times\frac{3}{12}}{\frac{9}{12}\times\frac{3}{12}+\frac{3}{12}\times\frac{2}{11}}$	**M1**	$\frac{9}{12}\times\frac{3}{12}=\frac{27}{144}$, $\frac{3}{16}$, 0.1875 seen as numerator or denominator of a fraction.
		M1	*Their* $\frac{9}{12}\times\frac{3}{12}+$ *their* $\left(\frac{3}{12}\times\frac{2}{11}\right)$ or *their* $\frac{27}{144}+$ *their* $\frac{1}{22}$ seen as denominator of a fraction. FT from part **(a)**.
	$\left[=\dfrac{\frac{3}{16}}{\frac{3}{16}+\frac{1}{22}}\right]=\frac{33}{41}$, 0.805	**A1**	Accept $\frac{4752}{5904}$ oe, 0.804878… rounded to at least three significant figures. If A0, **SC B1** for correct final answer www.
		3	

Question	Answer	Marks	Guidance
2(a)	**Method 1**		
	[P(X < 8) = 1 – P(8, 9, 10) =] 1 – ($^{10}C_8\,(0.7)^8\,(0.3)^2 + {}^{10}C_9\,(0.7)^9\,(0.3) + (0.7)^{10}$)	**M1**	One term $^{10}C_x\left(p\right)^x\left(1-p\right)^{10-x}$ with $0<p<1,\ x\neq 0$ or 10.
		A1	Correct unsimplified expression. Condone omission of last bracket only.
	= [1 – (0.2335 + 0.1211 + 0.0282)] = 0.617	**B1**	0.617 ⩽ p < 0.6175 www.
	Method 2		
	[P(0,1,2,3,4,5,6,7) =] $0.3^{10} + {}^{10}C_9\,0.7\,0.3^9 + \ldots + {}^{10}C_3\,0.7^7\,0.3^3$	**M1**	One term $^{10}C_x\left(p\right)^x\left(1-p\right)^{10-x}$ with $0<p<1,\ x\neq 0$ or 10.
		A1	Correct unsimplified expression.
	[= $5.905\times10^{-6}+1.378\times10^{-3}+\ldots+0.2668$] = 0.617	**B1**	0.617 ⩽ p < 0.6175 www.
		3	

Question	Answer	Marks	Guidance
2(b)	**Method 1**		
	[P(X < 5) =] $1-0.3^4$	**M1**	$1 - b^d$; b = 0.3, 0.7; d = 4, 5. $1 - c^e - (1 - c) \times c^{e-1}$; c = 0.3, 0.7; e = 5, 6.
	0.9919, $\frac{9919}{10000}$	**A1**	Condone 0.992. If M0 scored, **SC B1** for 0.9919 or $\frac{9919}{10000}$ only.
	Method 2		
	[P(X < 5) =] $(0.7) + (0.3)(0.7) + (0.3)^2(0.7) + (0.3)^3(0.7)$	**M1**	$(e) + (f)(e) + (f)^2(e) + (f)^3(e)$ [+ $(f)^4(e)$] e = 0.7, 0.3; $e + f = 1$.
	0.9919, $\frac{9919}{10000}$	**A1**	Condone 0.992. If M0 scored, **SC B1** for 0.9919 or $\frac{9919}{10000}$ only.
		2	
2(c)	$(0.4)^4\,(0.6)^2 \times 0.6 \times {}^6C_2$	**M1**	$(0.4)^4\,(0.6)^r$; r = 2, 3. No inappropriate addition.
		M1	$(0.4)^a\,(0.6)^b \times {}^6C_2$; $a + b$ = 6, 7.
	= 0.0829, $\frac{1296}{15625}$	**A1**	Accept 0.082944 correct to at least three significant figures. If A0 scored, **SC B1** for correct answer www.
		3	

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="5">3(a)</td><td><table><tr><td>Class width</td><td>20</td><td>10</td><td>5</td><td>5</td><td>10</td><td>20</td></tr><tr><td>Frequency density</td><td>0.4</td><td>2.3</td><td>7.0</td><td>10.4</td><td>2.0</td><td>0.6</td></tr></table></td><td>M1</td><td>At least five frequency densities (f/cw), e.g. $\frac{8}{20},\frac{23}{10},\ldots$
Accept unsimplified.</td></tr>
<tr><td rowspan="3">
</td><td>A1</td><td>All heights correct on graph (no FT).</td></tr>
<tr><td>B1</td><td>Bar ends at [0,] 20, 30, 35, 40, 50, 70 with a linear scale and at least three values indicated ‘linearly’.</td></tr>
<tr><td>B1</td><td>Axes labelled frequency density (fd) and time (mins).
Frequency density scale vertical starts at 0 with a linear scale and at least three values indicated ‘linearly’.
Axes can be reversed.</td></tr>
<tr><td></td><td>4</td><td></td></tr>
</table>

Question	Answer	Marks	Guidance
3(b)	Midpoints 10, 25, 32.5, 37.5, 45, 60	**B1**	At least five correct mid-points seen or used in formula.
	Mean = $\frac{10\times8+25\times23+32.5\times35+37.5\times52+45\times20+60\times12}{150}$ $\left[=\frac{5362.5}{150}\right]$	**M1**	Correct mean formula using their 6 midpoints (must be within class, not upper bound, lower bound). Condone one error. If correct midpoints seen, accept $\frac{5362.5}{150}$ or $\frac{80+575+1137.5+1950+900+720}{150}$.
	$= 35.75,\ 35\frac{3}{4}$	**A1**	Accept 35.8, not $\frac{143}{4}$. If A0 scored, **SC B1** for $35.75, 35\frac{3}{4}$ only.
		3	
3(c)	$30 \leqslant t < 35$	**B1**	Condone '3rd' interval, 30 – 35.
		1	

Question	Answer	Marks	Guidance
4(a)	[P(X > 1.26) = P$\left(Z >]\dfrac{1.26-1.20}{0.16}\right)$ $\left[=P\left(Z>0.375\right)\right]$	**M1**	Use of ± standardisation formula with 1.26, 1.20 and 0.16, not σ^2, not σ, no continuity correction.
	= 1 – 0.6462	**M1**	Calculating the appropriate probability area, (leading to their final probability, expect < 0.5).
	= 0.354	**A1**	0.3538, $0.3535 < p \leqslant 0.354$. Only dependent on the 2nd M mark. **SC B1** for $0.3535 < p \leqslant 0.354$ if M0M0A0 awarded.
	[500 × *their* 0.3538] = 176, 177	**B1 FT**	Strict FT *their* at least 4-figure calculated probability, seen anywhere (not a z-value). Final answer must be a single positive integer value, no approximation or rounding stated.
		4	
4(b)	$\left[\mathrm{P}\left(Z>\dfrac{2.40-2.50}{\sigma}\right)=0.20\right]$	**B1**	$-0.842 \leqslant z < -0.8415$ or $0.8415 < z \leqslant 0.842$ seen.
	$\dfrac{2.40-2.50}{\sigma}=-0.842$	**M1**	Use of the ±standardisation formula with 2.40, 2.50, σ and a z-value (not 0.20, 0.80, 0.158, 0.7881, 0.2119, 0.5793, 0.4207, …), not σ^2. Condone continuity correction of ±0.005. Condone $-\dfrac{0.1}{\sigma}=-0.842$ etc. for M1.
	$\sigma=0.119$	**A1**	$0.1185 < \sigma \leqslant 0.119$.
		3	

Question	Answer	Marks	Guidance
4(c)	[Mean = $80 \times 0.2 =$]16 [Variance = $80 \times 0.2 \times 0.8 =$]12.8	**B1**	16 and 12.8 seen, allow unsimplified. May be seen in standardisation formula. $\frac{8\sqrt{5}}{5}$, 3.5777… to at least three significant figures implies correct variance. Incorrect notation penalised.
	[P(X < 22) = P(Z <]$\frac{21.5-16}{\sqrt{12.8}}$) [P($Z < 1.537$) = $\Phi(1.537)$]	**M1**	Substituting *their* 16 (not 1.2, 2.5) and *their* 12.8 (not 0.16, *their* 0.119) in the ±standardising formula (any number for 21.5), condone σ^2 or $\sqrt{\sigma}$.
		M1	Using continuity correction 21.5 or 22.5 in *their* standardisation formula.
		M1	Appropriate area Φ, from final process, must be a probability.
	0.938	**A1**	$0.9375 < p \leqslant 0.938$.
		5	

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="3">5(a)</td><td>P(X = 2) = P(WLL or DDL)
= $0.5\times0.2^2\times{}^3\mathrm{C}_1+0.3^2\times0.2\times{}^3\mathrm{C}_1=\left[0.06+0.054\right]$</td><td>M1</td><td>$0.5\times0.2^2\times{}^3\mathrm{C}_a\left(or\,3\right)+x$; $0 < x < 1$; $a = 1, 2$.
Or
$0.3^2\times0.2\times{}^3\mathrm{C}_b\left(or\,3\right)$ $+ y$; $0 < y < 1$; $b = 1, 2$.
Or
$0.5\times0.2^2\times a+0.3^2\times0.2\times b$; $a, b = 1, 2, 3$.</td></tr>
<tr><td>$=0.114$</td><td>A1</td><td>AG.
Fully correct solutions with outcomes identified and linked to appropriate probabilities.
Condone 2 = W, 1 = D, 0 = L.
Probabilities alone do not identify outcomes.
If individual scenarios are identified, separate calculations must correspond to the order.</td></tr>
<tr><td></td><td>2</td><td></td></tr>
<tr><td rowspan="4">5(b)</td><td rowspan="2"><table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>P($X = x$)</td><td>0.008
$\frac{1}{125}$</td><td>0.036
$\frac{9}{250}$</td><td>0.114</td><td>0.207</td><td>0.285</td><td>0.225
$\frac{9}{40}$</td><td>0.125</td></tr></table></td><td>B1</td><td>One additional correct probability in table or clearly identified.</td></tr>
<tr><td>B1</td><td>A second additional correct probability in table or clearly identified.</td></tr>
<tr><td></td><td>B1</td><td>Final correct probability, all probabilities in table.
If 0/3 scored, SC B1 for three additional probabilities in table that sum to 0.269 exactly.</td></tr>
<tr><td></td><td>3</td><td></td></tr>
</table>

Question	Answer	Marks	Guidance
5(c)	[E(X) = $\left[0.008\times 0+\right]0.036\times 1+0.114\times 2+0.207\times 3+0.285\times 4$ $+0.225\times 5+0.125\times 6=$] [0] + 0.036 + 0.228 + 0.621 + 1.140 + 1.125 + 0.750 [= 3.9] OR E(X) = 3(0.5 × 2 + 0.3 × 1) [= 3.9]	**M1**	Accept unsimplified expression. May be calculated in the variance, FT *their* table with probabilities, $0 < p < 1$, that sum to 1. FT acceptable at the bold partially evaluated stage.
	[Var(X) = $\left[0.008\times 0^2+\right]0.036\times 1^2+0.114\times 2^2+0.207\times 3^2+0.285\times 4^2$ $+0.225\times 5^2+0.125\times 6^2-\textit{their}\,3.9^2=$] $\left[0.008\times 0+\right]0.036\times 1+0.114\times 4+0.207\times 9+0.285\times 16$ $+0.225\times 25+0.125\times 36-\textit{their}\,3.9^2$	**M1**	Appropriate variance formula using *their* $(\mathrm{E}(X))^2$ value. FT *their* table with probabilities, $0 < p < 1$, that may not sum to 1.
	$=\left[17.04-3.9^2\right]=1.83$	**A1**	Cao. Condone $\frac{183}{100}$.
		3	

Question	Answer	Marks	Guidance
6(a)	4M 1W: ${}^6C_4 \times {}^4C_1 = 60$	**M1**	${}^6C_a \times {}^4C_b$ with $a + b = 5$ seen, no extra terms.
	3M 2W: ${}^6C_3 \times {}^4C_2 = 120$ 2M 3W: ${}^6C_2 \times {}^4C_3 = 60$	**B1**	Correct outcome/value for one clearly identified scenario. Accept unsimplified, www. Condone use of $\times {}^5C_0$.
		M1	Add values of three correct scenarios, no incorrect scenarios, no repeated scenarios. Condone ${}^6C_a \times {}^4C_b$ with $a + b = 5$ to identify M, W for this mark.
	Total 240	**A1**	Not dependent on B1. If A0 scored, **SC B1** for 240 www.
		4	
6(b)	$6! \times 4! \times 2$	**M1**	$6! \times 4! \times k$; $k = 1, 2$. 1 can be implied.
	= 34 560	**A1**	Cao. If M0 scored, **SC B1** for 34 560 www.
		2	

Question	Answer	Marks	Guidance
6(c)	**Method 1** – Arrangements of OP in front row, 8 remaining people arranged.		
	$8! \times 3 \times 2$	**M1**	$8! \times g$, g an integer greater than 1.
		M1	$h! \times 3 \times j$; $h = 7, 8, 9$; $j = 1, 2$ (1 may be implied). Condone ${}^{3}C_{1}$ for 3.
		M1	$h! \times 3 \times 2$; $h = 7, 8, 9$. Condone ${}^{2}C_{1}$ for 2. (Condone $h! \times 3!$ For M1M1).
	= 241 920	**A1**	If A0 Scored, **SC B1** for 241 920.
	Method 2 – Two additional people selected for front row, front row arranged, remaining 6 people arranged in back row.		
	${}^{8}C_{2} \times 6! \times 3! \times 2$	**M1**	${}^{8}C_{a} \times d$, $a = 2,6$, d an integer greater than 1.
		M1	$6! \times e$, e an integer greater than 1.
		M1	${}^{8}C_{a} \times f! \times 3! \times 2$ or ${}^{8}C_{a} \times f! \times 6 \times 2$; $a = 2, 6$; $f = 5, 6, 7$.
	= 241 920	**A1**	If A0 Scored, **SCB1** for 241 920.
	Method 3 – Arrangements of two additional people for front row, front row arranged, remaining 6 people arranged in back row.		
	${}^{8}P_{2} \times 6! \times 3!$	**M1**	${}^{8}P_{2} \times d$, d an integer greater than 1.
		M1	$6! \times e$, e an integer greater than 1.
		M1	${}^{8}P_{2} \times h! \times 3!$ or ${}^{8}P_{2} \times h! \times 6$; $h = 5, 6, 7$.
	= 241 920	**A1**	If A0 Scored, **SC B1** for 241 920.

Question	Answer	Marks	Guidance
6(c)	**Method 4** – Arrangements of 6 people for back row, front row arranged.		
	$^8P_6 \times 3! \times 2!$	**M1**	$^8P_6 \times d$, d an integer greater than 1.
		M1	$3! \times e$, e an integer greater than 1.
		M1	$^8P_6 \times j! \times 2$; $j = 1, 2, 3$.
	= 241 920	**A1**	If A0 Scored, **SC B1** for 241 920.
		4	